

Two Students. Same Exam. Different Strategy.



Student A



Student B

- Both students want to **answer** questions correctly.
- But how they **prepare** makes all the difference.

👉 *This is exactly what happens in LLM systems too.*

Student A: Reads the Entire Book Every Time



Before every exam question...

- Opens the full textbook
- Searches all chapters
- Reads everything again
- Takes more time
- Gets slightly different **understanding** each time

✗ Slow **✗ Exhausting** **✗ Hard to scale before exams**



This is like RAG without cache

→ Retrieve + Generate every single time

Student B: Prepares Short Notes Once



- Creates **short notes**
- Stores key concepts
- Revises in minutes
- Reuses notes for **future exams**

✓ **Fast** ✓ **Efficient** ✓ **Reliable**

💡 **This is Cache-Augmented Generation (CAG)**

→ *Reuse previously computed knowledge*

Student Notes = LLM Cache

Student World	AI World
• Short notes	Cached answers
• Quick revision	Cache lookup
• Less effort	Low latency ⚡
• No re-reading	Lower cost 💰
• Same notes	Consistent answers 🎯

Smart systems don't redo work — they reuse it.

Why Cache-Augmented Generation Matters

- ⚡ **Faster** responses
- 💰 Lower LLM & retrieval cost
- 🎯 Consistent answers
- 📈 Scales to thousands of users
- 🚀 RAG gives knowledge
 - ⚡ Cache gives efficiency

The best systems use both.